



MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1 2025/2026

PROPOSAL MINI PROJECT

MICROCONTROLLER-BASED AUTOMATIC WASHING MACHINE WITH LOAD DETECTION

SECTION 2

GROUP 16

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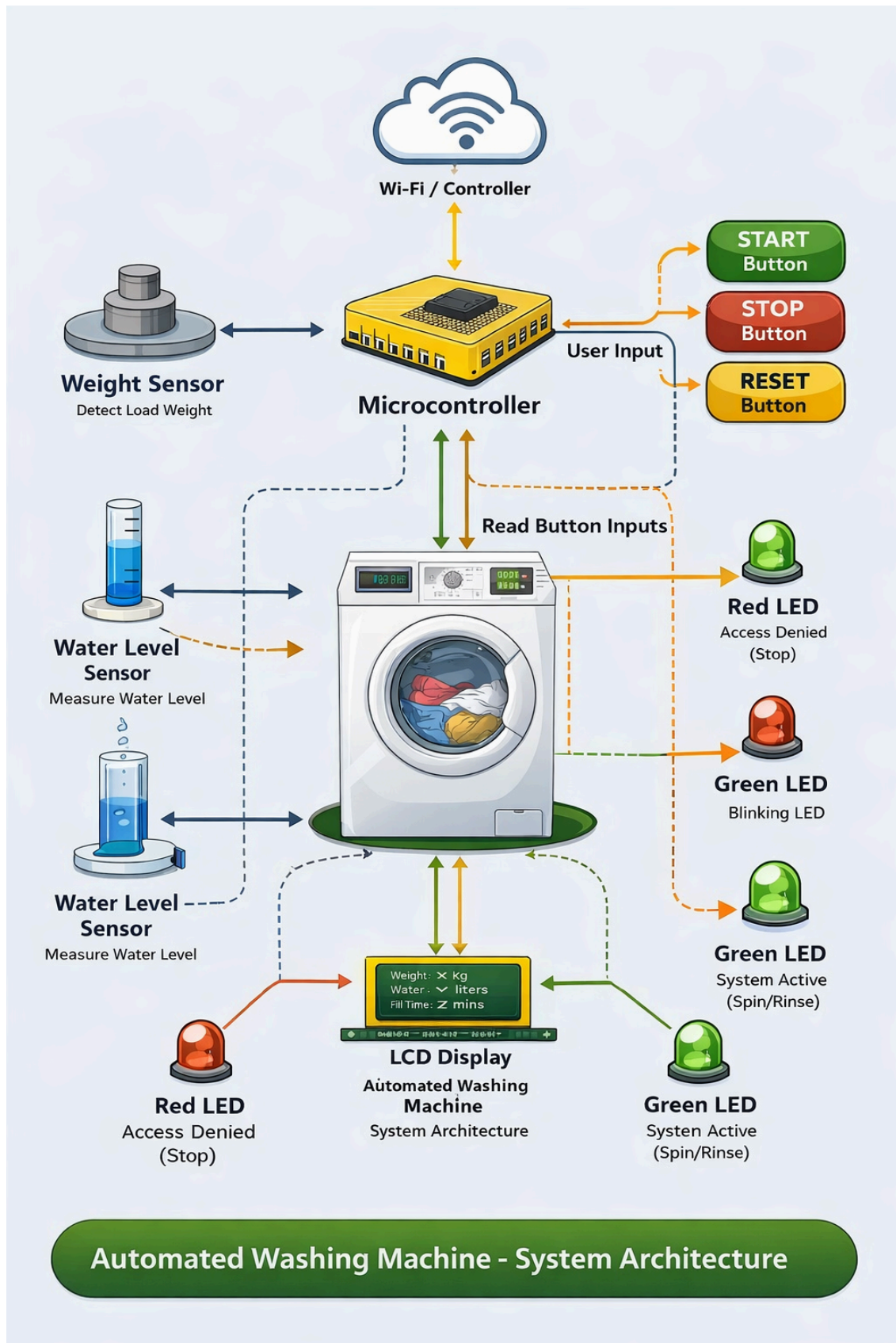
1. INTRODUCTION

The rapid advancement of automation and smart systems has significantly influenced the development of modern household appliances. One such appliance is the automatic washing machine, which integrates sensors, control systems, and user interfaces to improve efficiency, convenience, and safety. In line with the objectives of the Measurement, Sensors, and Instrumentation (MSI) course, this project proposes the design and development of a Smart Automated Washing Machine System that demonstrates the practical application of sensors, actuators, and microcontroller-based control.

This project focuses on automating the washing process by utilizing a weight sensor to detect the load of clothes and automatically determine the required water quantity and estimated processing time. An LCD display is used to provide real-time information to the user, while push buttons allow basic control such as start, stop, and reset functions. The washing, spinning, and rinsing processes are controlled using predefined timers, and LED indicators are implemented to clearly indicate the operational status of the motor.

Through this project, students are able to apply theoretical knowledge of sensors, timing control, and system integration into a functional prototype. The proposed system not only enhances user convenience but also promotes efficient water usage and clear process monitoring, making it a relevant and practical automation solution aligned with real-world applications.

2. SYSTEM ARCHITECTURE



3. WIRING DIAGRAM

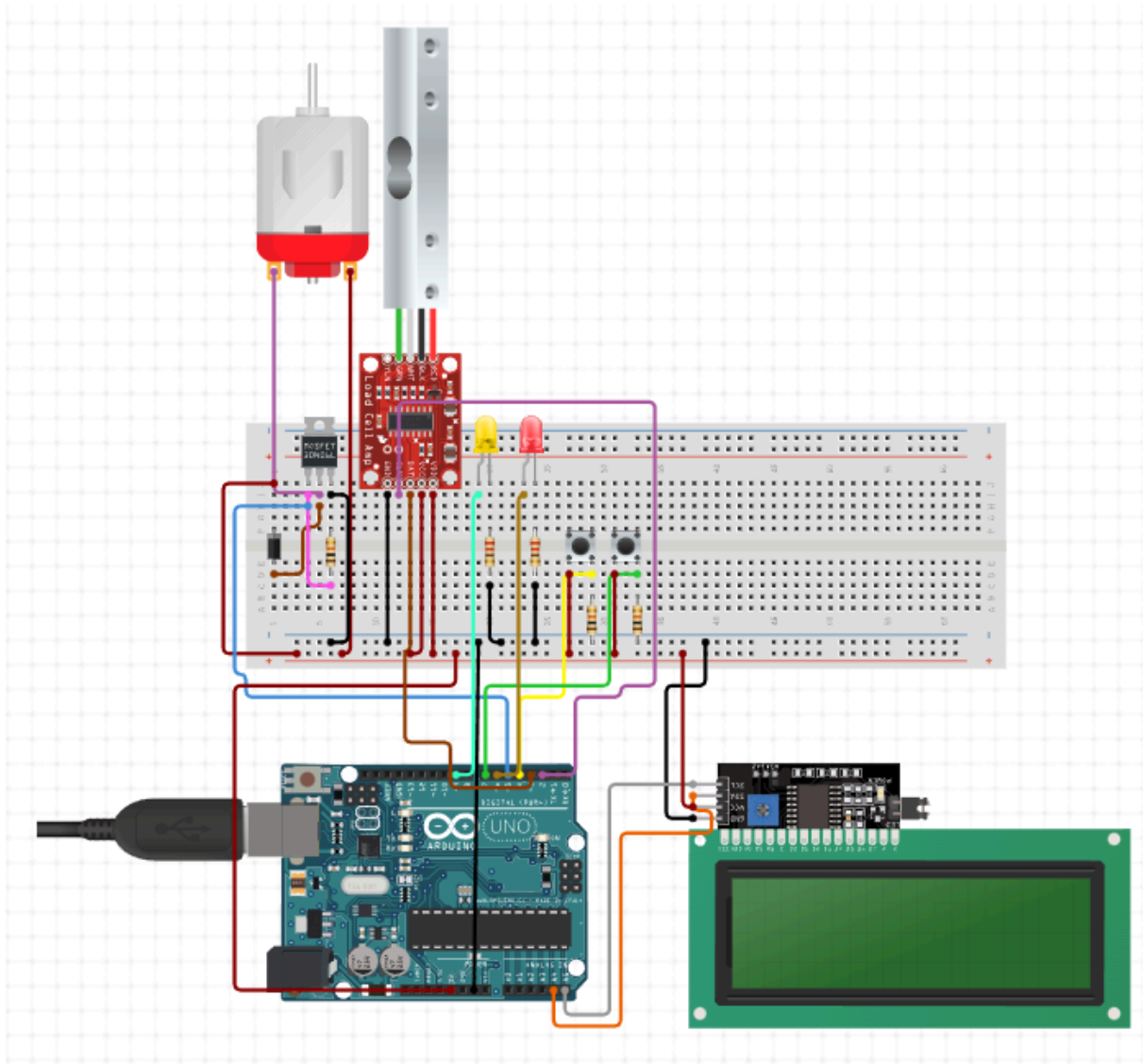


Figure 1: Electrical design

3.1 Key Hardware Breakdown

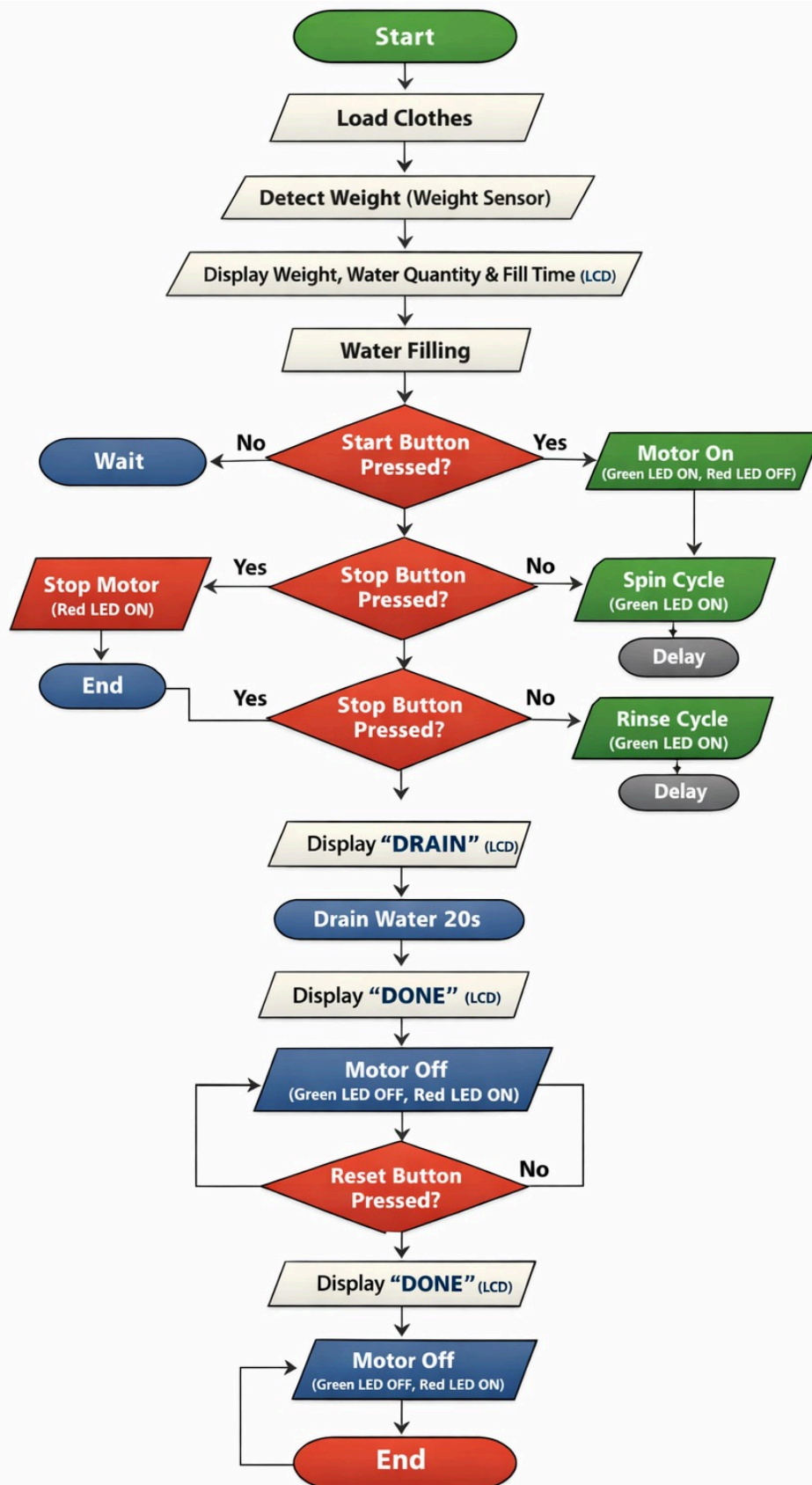
Based on the electrical diagram, the setup consists of three main systems:

- **The Controller (Arduino Uno):** The "brain" that manages the timing and processes inputs from the sensors.
- **The Weighing System (Load Cell & HX711):** Used to detect the total weight of the clothes to determine water quantity.
- **The Interface & Drive System:**
 - **I2C LCD Display:** Provides visual feedback.
 - **DC Motor & MOSFET:** Represents the washing machine drum, where the MOSFET acts as a high-power switch controlled by the Arduino.
 - **LEDs & Buttons:** Provide manual control and visual status of whether the motor is rotating.

3.2 Arduino Pin Assignment Table

NO	Component	Arduino Pin	Type	Purpose
1	HX711 DT	A0	Input	Weight Data
2	HX711 SCK	A1	Output	Weight Clock
3	LCD SDA	A4	I2C	Data Transfer
4	LCD SCL	A5	I2C	Clock Signal
5	Start Button	D2	Input	Begin Process
6	Stop Button	D3	Input	Reset/Halt
7	Green LED	D6	Output	Motor Active
8	Red LED	D7	Output	Machine Idle
9	MOSFET Gate	D9	Output	Motor PWM Control

4. FLOWCHART



5. REFERENCES

- Load Cell & HX711 Amplifier:** Avia Semiconductor. (n.d.). *HX711: 24-bit analog-to-digital converter (ADC) for weigh scales* [Data sheet].
https://www.mouser.com/datasheet/2/813/HX711_Datasheet-1522817.pdf
- LCD Display (I2C):** Arduino. (2016, March 7). *LiquidCrystal I2C library documentation*. <https://docs.arduino.cc/libraries/liquidcrystal-i2c/>
- Motor Control (MOSFET):** Instructables. (2023). *Speed controlling of DC motor using MOSFET and Arduino*.
<https://www.instructables.com/Speed-Controlling-of-DC-Motor-Using-MOSFET/>
- Automated Weighing & Control Logic:** Alyammahi, S., Alhmoudi, A., Alawadhi, M., & Alqaydi, F. (2025). Low-cost IoT-based smart grain monitoring system for sustainable storage management. *Engineering Proceedings*, 118(1), 90.
<https://doi.org/10.3390/ECSA-12-26545>
- Washing Machine Logic/Sequence:** Kaffale, T. H., & Liu, J. (2025). Speed and direction control of DC motor using Arduino Uno microcontroller. *Open Access Library Journal*, 12(3), 1-11. <https://doi.org/10.4236/oalib.1113007>

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been untaken or done by unspecified sources or persons. We hereby certify that this report has **not been done by only one individual, and all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have **read and understand** the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for **marking**, and this **final printed report** has been **verified by us**.

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